

True Multi-Tenant Migration Strategy

To migrate our disparate instances of the EHR software into a multi-tenant instance I would propose the following plan.

1. Ensure all tenants are running the most current version of the software. If not, remedy that first
2. Ensure we have key performance benchmarks already defined for primary user flows
3. Deploy the current version of the software to our new multi-tenant instance with no data.
4. Provision a shared database resource
5. Each schema will be unique to the tenant and the schema qualification would need to be injected into the queries based on user context
6. Establish a demo/test tenant in the new multi-tenant instances to vet connection patterns and plumbing, and provide us with an isolated production testbed for validating deployments
7. Identify our lowest utilization tenant to decrease the blast radius in the event of a failed migration.
8. Migrate the UI side of the identified low utilization tenant onto the new multi-tenant deployment, while maintaining a connection to the existing database.
9. If that goes well, incur downtime to backup and restore the single tenant database into the target schema in the new shared database resource identified above.
10. Monitor that migrated tenant for an agreed upon period of time to ensure that performance is not degraded compared to the single tenant instance benchmarks we collected
11. Once we are comfortable with this tenant, we would perform the same process for the remaining tenants ordered by tenant size or overall utilization from smallest to largest.